BANANI JANA

Jr. Geologist | Mineral Exploration | Mineral Resource Estimation | Remote Sensing & GIS

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PROFESSIONAL SUMMARY

Geologist (M.Sc. Applied Geology, 2026) working as Jr. Geologist at Shanvi Resources in mineral exploration and mineral resource estimation, with hands-on use of AutoCAD, Golden Software Strater, ArcGIS, QGIS and Global Mapper for borehole correlation, geological modelling and technical reporting. Trained at the Atomic Minerals Directorate in radiometric mapping and uranium-related exploration in the Singhbhum Shear Zone, with certifications from the Geological Survey of India and IIRS-ISRO in remote sensing and geophysics. Builds Python and machine-learning tools for orebody modelling, ore-potential zonation, ERT inversion and hydrogeochemical analysis. Open to relocation for exploration, mining and geospatial roles.

PROFESSIONAL EXPERIENCE

Jr. Geologist Aug 2026 – Present

Shanvi Resources, Bhubaneswar, Odisha | Full-time | On-site

- Work on mineral exploration and mineral resource estimation, including borehole correlation and geological report writing.
- Prepare geological sections and drawings in AutoCAD and Golden Software Strater, and carry out spatial analysis and mapping in ArcGIS, QGIS and Global Mapper.
- Support project management and technical reporting for exploration work.

TRAINING & FIELD EXPERIENCE

Industrial training: Atomic Minerals Directorate for Exploration & Research, Ghatshila, Jharkhand (4–14 Aug 2025): geological mapping and radiometric checking in the Surda–Terenga area of the Singhbhum Shear Zone, with exposure to ore exploration and mapping, and to lithology and radiation-anomaly identification. Visited the Uranium Corporation of India Limited (UCIL) Narwapahar underground mine, with basic exposure to underground mining operations and mine planning.

Field work: Structural mapping and tectonic study along the North Purulia Shear Zone (Mar 2025); water quality measurement, bathymetric surveying and biomass estimation at Taki with Adamas University (Aug 2025), using an echo sounder, Abney level, multiparameter probe, laser distance finder, GPS and ArcGIS; open-cast coal mine, clay mine and dam visit at Kharia (2024); litho-structural mapping of the Chaibasa Formation at Galudih (2023); petrological studies at Maithon (2022).

Earlier work: Gramin Dak Sevak, Department of Posts, Government of India, Hooghly, West Bengal (Jan 2022 – Jan 2025).

TECHNICAL SKILLS

Exploration & geology	Mineral resource estimation, borehole correlation, core logging, radioactivity mapping, geological report writing, orebody modelling, sample and thin-section preparation, petrographic microscopy, grain-size analysis
GIS & remote sensing	ArcGIS, QGIS, Global Mapper, Google Earth Pro; GRACE TWSA, water body monitoring, water storage mapping
Geophysics & hydrogeology	ERT inversion (pyGIMLi), VES, MODFLOW/MODPATH (GMS), HEC-HMS, PHREEQC, statistical modelling
Data & machine learning	Python (Pandas, NumPy, Scikit-learn, Matplotlib), machine learning, Random Forest, data visualization
Software & instruments	AutoCAD, LibreCAD, Golden Software Strater, Surfer, RockWorks, Geo-Orient, Geo-Rose, CorelDRAW, MS Office; GPS (DNRGarmin), echo sounder, Abney level, laser distance finder

EDUCATION

M.Sc. Applied Geology Sep 2024 – Jul 2026

Presidency University, Kolkata | CGPA 8.65 / 10

Dissertation (2026): Integrated Hydrogeological and Geochemical Investigation of River-Aquifer Interaction in Parts of Hooghly–Rupnarayan Interfluve: A Study from Lower Gangetic Plain, India. Supervisor: Dr. Aditya Sarkar.

B.Sc. (Hons) Geology Nov 2021 – Jun 2024

Jogamaya Devi College, University of Calcutta | CGPA 7.5 / 10

National Scholarship Portal (NSP) Scholar, Government of India (2021); Rank 3, PUMBDET postgraduate entrance examination, Presidency University (2024).

SELECTED PROJECTS

- **Orebody geometry modelling and grade-tonnage computation from drillhole data** (Python) – GitHub
- **Data-driven ore potential zonation from multi-criteria geospatial evidence** (machine learning) – GitHub
- **ERT-based subsurface resistivity characterisation** (Python, pyGIMLi) – GitHub
- **Subsurface characterisation using well log petrophysics and seismic forward modelling** (Python) – GitHub
- **Hydrogeochemical data processing and statistical modelling** (Python) – GitHub
- **Random Forest classification of groundwater hydrochemical facies from water-quality parameters** (scikit-learn) – GitHub
- **Environmental impact assessment of mining-zone water quality** (geochemical indices, ML), collaborative – GitHub
- **Machine-learning-assisted Groundwater Resilience Index (GRI)** (ML), collaborative – GitHub

PUBLICATIONS & CONFERENCES

- **Evaluating Groundwater Resilience for WASH Planning in a Semi-Arid Region Through a GIS-Machine Learning Framework: Alwar District, Rajasthan** – National WASH Conference, SPM-NIWAS, 2026 (first author)
- **Machine Learning-Enhanced Analysis of Temporal b-Value Variations Across Different Tectonic Settings** – BMES-2026, IIT (ISM) Dhanbad, 2026 (first author)
- **Integrated Morphotectonic Assessment of the Torsa–Amo Chu River Basin Using Geomorphic Indices and Machine Learning** – BMES-2026, IIT (ISM) Dhanbad, 2026
- **Integrated Hydro-Geophysical and GIS-Based Characterization of Micro-Level Aquifers in the Sundarbans, West Bengal** – Water Summit 2025, Manav Rachna University (first author)
- **Forecasting Groundwater Dynamics in Jalpaiguri District, North Bengal, Using Machine Learning and GRACE Satellite Data** – Water Summit 2025, Manav Rachna University
- **Quantification of Groundwater Quality Issues in Reference to Urbanization in Parts of Barddhaman District, West Bengal** – 32nd West Bengal State Science & Technology Congress, 2024
- **Carbon Dioxide Sequestration and Role of Basaltic Minerals in It** – Archipelago '24, Ballygunge Science College, University of Calcutta, 2024 (first author)

CERTIFICATIONS & WORKSHOPS

- **Geological Survey of India (GSITI):** Application of Remote Sensing in Mineral Exploration; Regional Geophysical Mapping for Mineral Targeting; Workshop on Interpretation of Different Kinds of Geophysical Datasets with Case Studies (2025)
- **IIRS, ISRO:** Remote Sensing and Digital Image Analysis (2025); Advances in Remote Sensing Techniques for Geological Applications (2025); Advanced Image Analysis for Geospatial Professionals (2025)
- **University of Oulu, Finland:** Application of Remote Sensing in Water Resources and Climate Change, PhD short course, 2 ECTS (2025)
- **GIS:** ArcGIS Imagery MOOC, Esri (2025); QGIS Mastery, Quest GIS; Application of Remote Sensing and GIS in Geosciences, Jogamaya Devi College (2024)
- **Python and ML:** Machine Learning I, Columbia University; Applied Data Science with Python, IBM (2025); Python Programming in Geology, TerraPy (2025)
- **IIT (ISM) Dhanbad:** Workshop: Bridging Mathematics and Earth Sciences, BMES-2026